



Acoustic printing of conductive polymers

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Fabricating materials within optically opaque structures, such as biological tissue, is a considerable challenge. Recently, ultrasound-based printing (“sonoprinting”) approaches have emerged as a promising strategy to address this challenge. However, an approach to sonoprint conductive materials has yet to be realized, limiting potential bioelectronic applications. Here, we extend sonoprinting to conductive materials by designing temperature-based and pressure-based methods to polymerize conductive polymers with focused ultrasound (FUS). Our temperature-based approach relies on the acoustic attenuation of the surrounding medium to generate heat under FUS, whereas our pressure-based approach leverages the acoustic vaporization of perfluorohexane double emulsions to trigger polymerization. We demonstrate that both approaches can be used to print the conductive polymer poly(3,4-ethylenedioxythiophene) (PEDOT) through optically opaque hydrogels and biological tissue with high spatial resolution. Taken together, our results establish complementary temperature- and pressure-based methods for sonoprinting conductive polymers, paving the way for future efforts to fabricate bioelectronic interfaces in tissue.

acoustic printing | sonoprinting | conductive polymers | bioelectronics

Additive manufacturing is emerging as a powerful approach for fabricating electroactive materials (1) with diverse applications in tissue engineering (2, 3), neurophysiology (4, 5), biosensing (6–8), and soft robotics (9). Although conventional additive manufacturing techniques, such as photopolymerization and extrusion-based methods (10), are widely used, these approaches rely on optical or physical access to the print environment, limiting their application in biological tissue *in situ*. Specifically, photopolymerization is constrained by scattering and absorption events that limit the penetration depth of light in tissue (11, 12), whereas extrusion-based methods require invasive surgical approaches to access deep tissue (13).

Focused ultrasound (FUS) is a promising alternative to light for delivering energy to trigger polymerization due to its improved penetration depth in optically opaque materials (14, 15). Several methods for FUS-based printing (“sonoprinting”) have recently been proposed (14, 16), including a cavitation-based approach to cure a polydimethylsiloxane (PDMS) resin (16) and a temperature-based approach to polymerize poly(ethylene glycol) diacrylate (PEGDA) with the thermal initiator ammonium persulfate (APS) (14). However, a method to sonoprint electroactive materials, such as conductive polymers, has yet to be realized, limiting the utility of sonoprinting for bioelectronic applications (15).

To address this challenge, here, we introduce two methods for sonoprinting conductive polymers using distinct temperature-based and pressure-based mechanisms. The temperature-based sonoprinting approach leverages the acoustic attenuation of the surrounding medium to generate localized and transient heat at the FUS focus, decomposing APS to polymerize the conductive polymer poly(3,4-ethylenedioxythiophene) (PEDOT). The pressure-based approach relies on acoustic droplet vaporization of perfluorohexane double emulsions to combine an initiator (APS), monomer [4-((2,3-dihydrothieno[3,4-b][1,4]dioxin-2-yl)methoxy)butane-1-sulfonate, EDOT-S] and catalyst [Fe(II)] to polymerize the PEDOT-derivative PEDOT-S (17). We demonstrate that these methods can be used to sonoprint conductive polymer structures in aqueous solutions, hydrogels, and deep tissues with a spatial resolution down to ~300–500 μm . We further analyze the temperature dependence of the sonoprinting reaction mechanism and show that our pressure-based approach operates in a biocompatible temperature regime (18) in contrast to previous sonoprinting approaches (14, 16).

Significance

The ability to fabricate materials in biological tissue could enable the development of minimally invasive methods to repair bones, monitor heart function, and stimulate neurons. Recent studies have proposed using focused ultrasound (FUS) to achieve this goal because acoustic energy can travel through tissue to deliver energy for printing. However, existing acoustic printing approaches have been restricted to nonconductive materials, preventing their use in bioelectronic interfaces for monitoring or stimulating tissue. Here, we help address this challenge by introducing two approaches to print conductive polymers with FUS.

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Results

Temperature-Based Approach to Sonoprint Conductive Polymers.

We designed a temperature-based sonicated ink ("sonoink") for printing conductive polymers, inspired by recent work on sonoprinting PEGDA hydrogels (14). We focused specifically on the conductive polymer PEDOT due to its biocompatibility (19) and widespread use in bioelectronics (4, 20). Our temperature-based sonoink contains the monomer 3,4-ethylenedioxythiophene (EDOT) and thermal initiator ammonium persulfate (APS) dispersed in a medium with high acoustic attenuation, such as a hydrogel or biological tissue (21). When a pressure wave from FUS passes through the acoustically attenuating medium, the acoustic energy is transformed into heat, decomposing APS into sulfate radicals to initiate the PEDOT polymerization reaction (Scheme 1).

Temperature-Based PEDOT Sonoprinting in Hydrogels and Tissue.

To illustrate our temperature-based sonoprinting approach, we prepared an alginate hydrogel containing the temperature-based sonoink. We placed the alginate hydrogel on a stage in an FUS printing setup (Fig. 1A), then applied 5 MHz FUS with a peak pressure of 3.6 MPa and high duty cycle of 90% to the hydrogel. After 10 min of FUS, we observed the formation of a PEDOT dot with a diameter of 1.4 mm through the optically opaque hydrogel (Fig. 1B and C). We validated the formation of PEDOT following treatment of the temperature-based sonoink with FUS using ^1H NMR (SI Appendix, Figs. S1 and S2 and Materials and Methods). To confirm that our sonoprinting approach generalizes to other media, we also prepared a polyvinyl alcohol (PVA) hydrogel with the temperature-based sonoink and verified that we could successfully sonoprint a PEDOT dot in the PVA hydrogel (SI Appendix, Fig. S3A).

We then optimized the spatial resolution of our temperature-based sonoprinting approach by minimizing the total duration of FUS. In a PVA hydrogel containing the sonoink, we observed that small PEDOT dots (mean diameter = 305 μm) could be printed after ~30 s of FUS (SI Appendix, Fig. S3). This spatial resolution is consistent with the focal diameter of the 5 MHz transducer used here (−6.0 dB focal diameter = 425 μm), suggesting that the FUS focal diameter determines the achievable spatial resolution. The spatial precision of our temperature-based sonoprinting approach allowed us to print detailed PEDOT patterns, such as a square, triangle, and Stanford logo, by programming a motorized translation stage to move the transducer while printing (Fig. 1D–F). This approach can also be extended to sonoprint 3D patterns by moving the transducer in the z-axis, but the achievable axial resolution is limited by the large axial beamwidth of the 5 MHz transducer used here (−6.0 dB axial beamwidth = 3.34 mm) (SI Appendix, Fig. S4).

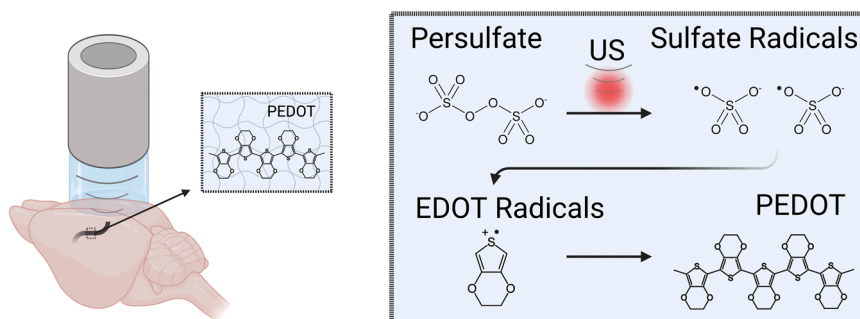
Finally, we tested whether our temperature-based sonoprinting approach could print PEDOT through thick, optically opaque biological tissues. Specifically, we prepared alginate hydrogels containing the temperature-based sonoink on top of a slice of chicken breast (thickness = 7.2 mm) and inside a mouse skull (combined thickness = 5.0 mm), then applied FUS. We printed PEDOT dots through both the chicken breast and skull samples (SI Appendix, Fig. S5), thus confirming that our approach enables deep penetration printing in biological tissue.

Temperature Dependence of the PEDOT Sonoprinting Reaction.

To better understand the dependence of the PEDOT sonoprinting reaction on the temperature changes induced by FUS, we conducted a series of kinetics experiments with the sonoink in a temperature-controlled aqueous environment.

First, we measured the UV-vis absorbance spectrum over the course of the reaction of EDOT and APS at 40 °C in an aqueous solution. We observed the formation of a shoulder absorbance band at 320 nm, followed by an absorbance peak at 800 nm (Fig. 2A). The 320 nm shoulder corresponds to the formation of EDOT dimers (22, 23), while the 800 nm peak arises from PEDOT and its extended conjugated π -system (17, 24, 25).

We then varied the temperature, concentration of APS, or concentration of EDOT while holding other reaction conditions constant and measured the UV-vis absorbance spectrum over the course of the reaction (Fig. 2B and SI Appendix, Figs. S6 and S7). We observed that the rate of change of absorbance at 320 nm and 800 nm increased with increasing temperature (Fig. 2B). More interestingly, the absorbance at 320 nm was a linear function of time at the onset of the reaction, whereas the absorbance at 800 nm was a nonlinear function of time in all temperature conditions examined (Fig. 2B). To better understand this result, we modeled the overall reaction as a two-step process consisting of a dimerization step where EDOT and APS react to form EDOT dimers, followed by a polymerization step where EDOT dimers and APS react to form PEDOT (SI Appendix, Supplementary Text). Adopting this model to analyze our kinetics data, we found that APS and EDOT are both first-order in the dimerization reaction (SI Appendix, Fig. S8A and B and Supplementary Text). Moreover, the temperature dependence of the dimerization reaction is well described by the Arrhenius equation with an activation energy of 34 kJ/mol (SI Appendix, Fig. S8C and Supplementary Text). In the polymerization step, we found that the reaction order of the EDOT dimer is dependent on the initial reaction condition [i.e., temperature, [EDOT], and [APS]] (SI Appendix, Fig. S9). Specifically, the reaction order of the dimer increases with increasing EDOT concentrations and is highest at intermediate temperatures and APS concentrations [40 °C and [APS] = 88 mM, respectively]. This suggests that the polymerization reaction



Scheme 1. Temperature-based sonoprinting approach leverages the acoustic attenuation of the surrounding medium (e.g., hydrogel or brain tissue) to generate transient and localized heat under focused ultrasound (FUS) and decompose the thermal initiator ammonium persulfate to polymerize EDOT. Created in BioRender.

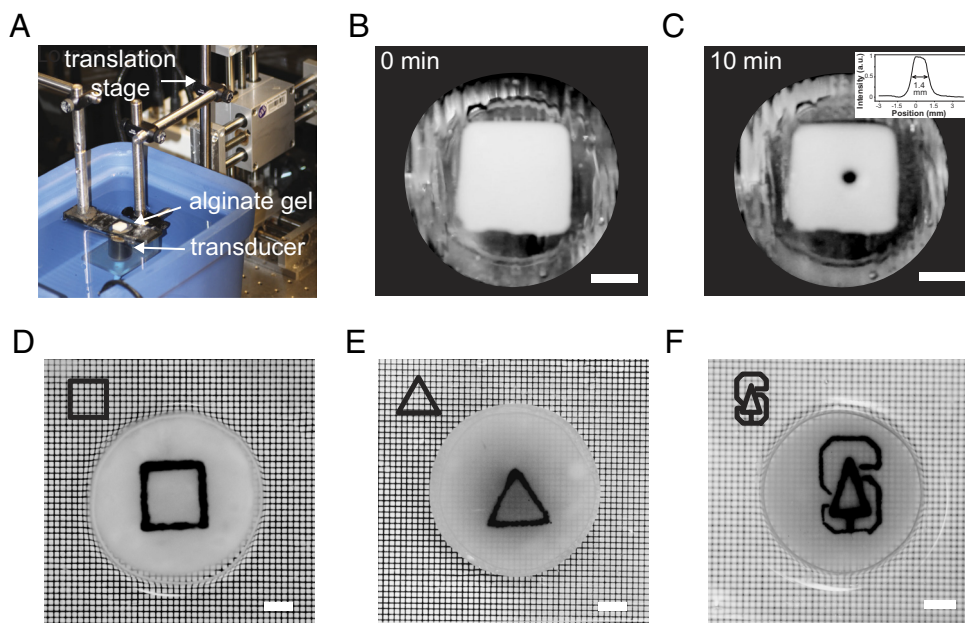


Fig. 1. Printing PEDOT with FUS using a temperature-based sonoprinting approach. (A) FUS printing setup. An alginate hydrogel sample is placed at the focus of an FUS transducer controlled with a motorized translation stage. (B and C) An alginate hydrogel with sonoink components above the FUS transducer before (B) and after (C) applying FUS for 10 min to print a PEDOT dot with a diameter of 1.4 mm. The inset shows the line intensity profile across the center of the printed PEDOT dot. (D–F) PEDOT square (D), triangle (E), and Stanford logo (F) printed with FUS. Insets in the *Upper Left* denote the input patterns for printing. FUS parameters are 5 MHz, 3.6 MPa, 90% duty cycle, and 1 s cycle length in panels C–F. (Scale bar, 5 mm in panels B–F.)

proceeds via different mechanisms [e.g., step-growth vs chain-growth polymerization (26)] depending on the initial reaction conditions, making further analysis of the kinetics of the polymerization step challenging. Nonetheless, our observation that the dimerization reaction is an Arrhenius process is consistent with a prior temperature-based sonoprinting study that used APS as the thermal initiator (14).

These results suggest that the PEDOT sonoprinting reaction is highly temperature dependent. Consequently, the formation of PEDOT patterns during sonoprinting should closely follow the spatial temperature profile induced by FUS. To verify this relationship, we measured the temperature of an alginate hydrogel containing the temperature-based sonoink using a thermal camera while applying FUS. The temperature reached $\sim 75^{\circ}\text{C}$ at the focus and declined sharply within a few millimeters (Fig. 2C). The gray-scale intensity of the PEDOT dot that formed after applying FUS for 2 min mirrored the corresponding temperature profile with the darkest color at the region with the highest temperature of $\sim 75^{\circ}\text{C}$, a lighter color in the neighboring ~ 60 to 70°C region, and no color change in the ~ 20 to 50°C surround (Fig. 2D). We compared these results to polymerization under the same heating conditions without FUS by preparing an alginate hydrogel with the temperature-based sonoink and placing it in an 80°C oven. The hydrogel turned dark black, characteristic of PEDOT formation, whereas an identical control hydrogel left at room temperature remained pale white (*SI Appendix, Fig. S10*). Taken together, these results suggest that the temperature-based sonoprinting reaction is an Arrhenius process and confirm the proposed sonoprinting mechanism (Scheme 1).

The temperature-based sonoprinting approach could be valuable for materials science applications that require printing conductive materials in optically opaque media incompatible with photopolymerization (15). However, as with prior sonoprinting methods (14–16), it is not suitable for bioelectronic applications in living tissue due to the high temperatures required. Nonetheless, our results thus far demonstrate that FUS can be leveraged to print

PEDOT structures with a spatial resolution of $\sim 300\ \mu\text{m}$ through thick overlying tissue.

Pressure-Based Approach to Sonoprint Conductive Polymers.

To enable sonoprinting conductive polymers at biocompatible temperatures, we designed a pressure-based sonoink to print the PEDOT derivative PEDOT-S (17). We focused on PEDOT-S here because it can be cross-linked by physiological levels of divalent cations, a feature that makes it a promising candidate for future in vivo bioelectronic applications (17, 27).

Our pressure-based sonoink relies on water-in-perfluorohexane-in-water double emulsions to trigger polymerization under FUS, inspired by recent work on FUS-triggered drug delivery (28–30). More specifically, the sonoink incorporates perfluorohexane double emulsions that separate an initiator (APS) from a monomer (EDOT-S) and catalyst (Iron (II)). FUS vaporizes the perfluorohexane shell via acoustic droplet vaporization (31), releasing APS to react with Iron (II). Iron (II) rapidly decomposes APS into sulfate radicals at physiological temperatures, initiating the PEDOT-S polymerization reaction (Scheme 2).

Double Emulsion Generation and Characterization for Pressure-Based Sonoprinting.

To realize the pressure-based sonoprinting strategy, we fabricated a microfluidic device for double emulsion generation (Fig. 3A) (32). We then generated perfluorohexane double emulsions containing APS by flowing the inner aqueous, oil, and outer aqueous solutions through the device (Fig. 3B and *Movie S1* and *Materials and Methods*). We visualized the resulting emulsions with an optical microscope and observed stable emulsions with a circular core surrounded by a shell, thus confirming successful double emulsion generation (Fig. 3C). Finally, we characterized the size and uniformity of the double emulsions by quantifying the inner and outer diameters of a batch of droplets. The double emulsions were highly uniform, with a median inner diameter of $31\ \mu\text{m}$ and outer diameter of $44\ \mu\text{m}$ (Fig. 3D).

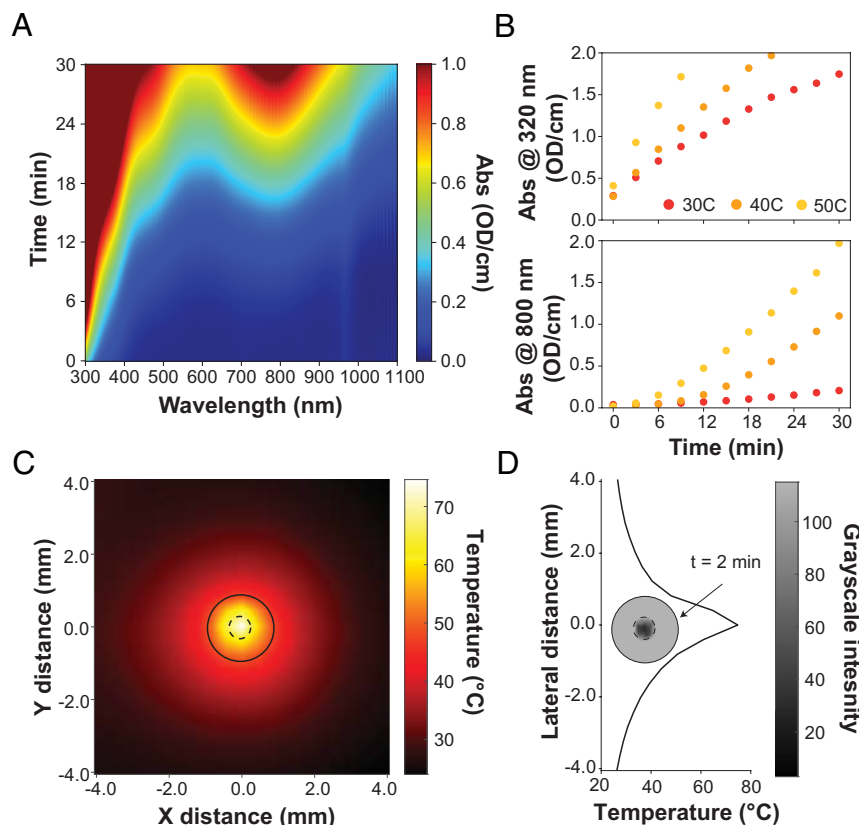
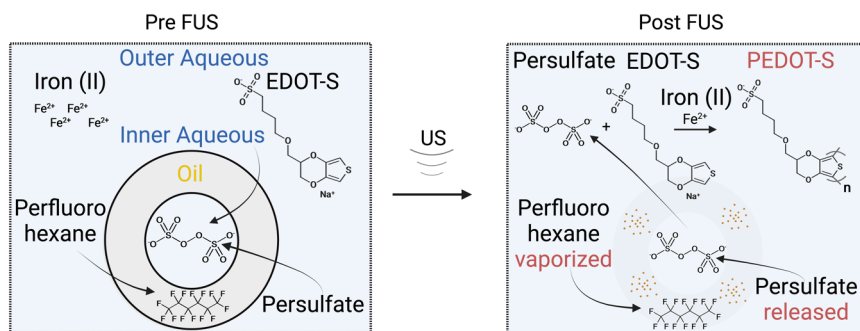


Fig. 2. Characterization of the kinetics of the temperature-based PEDOT sonoprinting approach. (A) Time-dependent UV-Vis measurements of the absorbance spectrum over the course of the reaction between EDOT and APS at 40 °C in an aqueous solution. A shoulder absorbance band at 320 nm and an absorbance peak at 800 nm appear during the reaction. (B) Absorbance at 320 nm and 800 nm over the course of the reaction at 30, 40, and 50 °C in an aqueous solution. (C and D) Temperature profile (C) and temperature cross-section at the FUS focus (D) in an alginate hydrogel while applying 5 MHz FUS with 3.6 MPa, 90% duty cycle, and 1 s cycle length. Inset in (D) shows the grayscale image of the PEDOT dot printed in the hydrogel by applying FUS for 2 min. The dashed black circle illustrates the size of the printed dot in (C and D), and the solid black circle in (C) illustrates the outline of the corresponding grayscale image in (D).

Pressure-Based PEDOT-S Sonoprinting in Aqueous Solutions, Hydrogels, and Tissue. To illustrate our pressure-based sonoprinting approach, we prepared an aqueous solution containing the pressure-based sonoink. In contrast to the temperature-based sonoink, the pressure-based sonoink should be polymerizable in aqueous solutions as it does not rely on the attenuation of FUS by the surrounding medium. We applied 5 MHz FUS to the pressure-based sonoink with a low duty cycle of 1%. After 10 min of FUS, the solution turned dark black, characteristic of PEDOT-S formation, whereas an identical control solution without FUS remained pale white (Fig. 4A). The UV-vis absorbance spectrum of the polymerized

product had a broad absorbance shoulder in the near infrared region (Fig. 4B), consistent with previous studies on PEDOT-S (17) and a positive control experiment where we polymerized PEDOT-S without ultrasound (*SI Appendix, Fig. S11*). Moreover, ^1H NMR of the pressure-based sonoink treated with FUS revealed spectral features consistent with the formation of PEDOT-S (*SI Appendix, Fig. S12* and *Materials and Methods*). Temperature measurements of the solution at the focus revealed negligible temperature increases while applying FUS (Fig. 4C), thus confirming that our approach relies on a pressure-based rather than a temperature-based mechanism.



Scheme 2. Pressure-based sonoprinting approach leverages acoustically responsive perfluorohexane double emulsions to separate an initiator (persulfate) from the monomer (EDOT-S) and catalyst [Iron (II)]. Applying FUS vaporizes the perfluorohexane shell via acoustic droplet vaporization, releasing persulfate to initiate the reaction and polymerize EDOT-S. Created in BioRender.

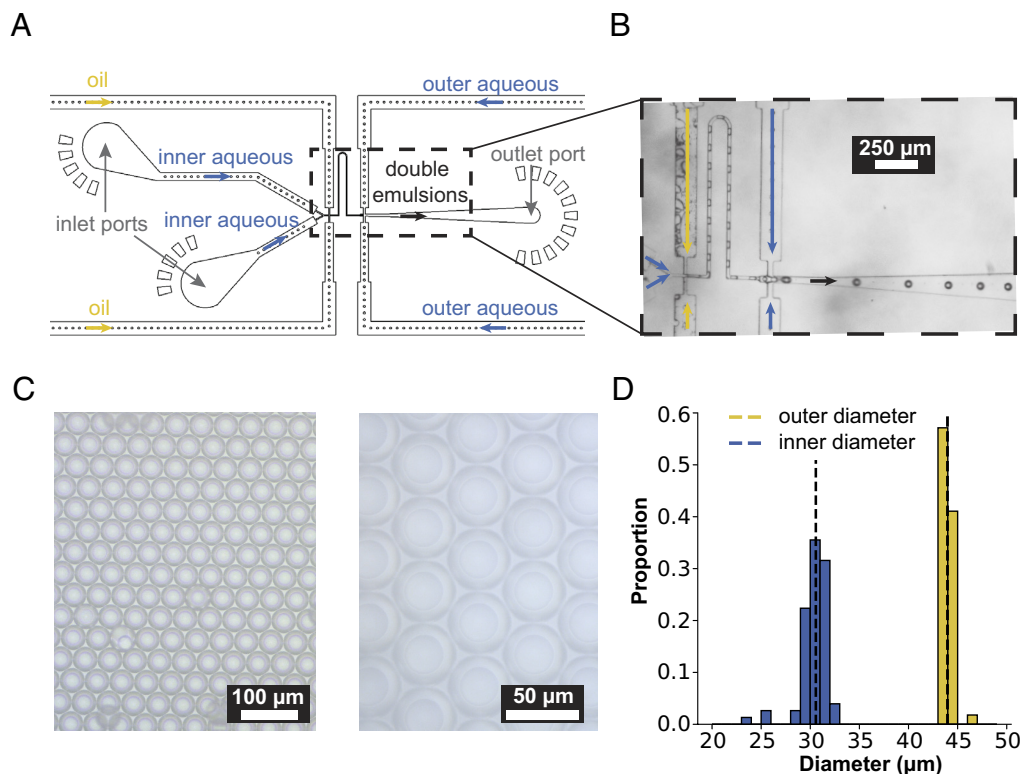


Fig. 3. Production of acoustically responsive perfluorohexane double emulsions. (A) Schematic of device used for double emulsion generation. Arrows indicate direction of flow of oil, inner aqueous, outer aqueous, and double emulsion solutions. (B) Image of device during double emulsion generation. (C) Representative optical microscope images of resulting double emulsions at two different magnifications. (D) Distribution of diameters of the inner aqueous core (inner diameter) and perfluorohexane shell (outer diameter) of double emulsions.

In an agarose hydrogel preparation, we found that small PEDOT-S dots with an average diameter of $509 \pm 27 \mu\text{m}$ could be printed with 5 MHz FUS (*SI Appendix, Fig. S13*). Moreover, we observed that the pressure-based sonoprinting approach was compatible with FUS frequencies ranging from 650 kHz to 5 MHz, with higher FUS frequencies associated with smaller printed dots (*SI Appendix, Fig. S14*). Notably, lower FUS frequencies are useful for applications that require sonoprinting at deeper locations and over larger areas. Furthermore, we evaluated the stability of a FUS-printed dot in an agarose hydrogel over an extended period. We found that the dot remained stable over two months, as indicated by a consistent diameter, suggesting the printed PEDOT-S does not diffuse or react with the surrounding medium and no additional PEDOT-S formed elsewhere in the hydrogel in the absence of ultrasound (*SI Appendix, Fig. S15*).

Importantly, the maximum pressure used in our experiments with the 5 MHz transducer was 3.6 MPa, corresponding to a mechanical index (MI) of 1.6. This MI is below the safety limits recommended by the FDA for diagnostic ultrasound (maximum MI = 1.9) and therefore falls well within the range considered acceptable for biomedical use. Moreover, we found that pressures as low as 2.3 MPa (corresponding to an MI of 1.0) are sufficient to induce noticeable color changes that suggest successful production of PEDOT at the ultrasound focus (*SI Appendix, Fig. S16*), suggesting pressure could be further reduced if necessary for particular applications.

Having established that our pressure-based approach enables sonoprinting PEDOT-S at biocompatible temperatures, we next sought to determine the conductivity of sonoprinted PEDOT-S. To do this, we prepared films with either PEDOT-S sonoprinted with 650 kHz FUS or a commercial PEDOT sample, and measured their conductivity with a four-point probe. The sonoprinted PEDOT-S film exhibited slightly higher conductivity (0.16 S/cm)

than the commercial PEDOT:PSS film (0.08 S/cm), consistent with prior reports that have attributed the enhanced conductivity of PEDOT-S to the absence of insulating PSS (33) (*SI Appendix, Fig. S17*).

These results suggest that it should be possible to print conductive PEDOT-S through tissue at biocompatible temperatures and pressures using our pressure-based sonoink. To verify this, we placed our pressure-based sonoink atop a 7.6 mm thick piece of porcine muscle (*Fig. 4D*), then applied FUS with a 1% duty cycle for 10 min. We observed the gradual formation of a PEDOT-S dot through the thick, optically opaque tissue (*Fig. 4E* and *Movie S2*). Temperature measurements of the tissue at the focus revealed negligible temperature increases while applying FUS (*Fig. 4F*), despite the high acoustic attenuation of muscle tissue (21), due to the low 1% duty cycle used here. To confirm that our approach can also be used in neural tissue, a domain of particular interest for future bioelectronic applications (4, 19), we repeated this experiment with a mouse skull and a 7.0 mm thick slice of sheep brain and found that we could successfully print a PEDOT-S dot through both the skull and thick brain tissue (*SI Appendix, Figs. S18 and S19*). Finally, we applied FUS to the pressure-based sonoink on a mouse brain in vivo (*Fig. 4G*) to print a PEDOT-S dot atop the anterior left hemisphere (*Fig. 4H* and *I*). This result provides a proof-of-principle for extending the pressure-based sonoprinting approach to in vivo settings. Taken together, these results demonstrate that our pressure-based sonoink can be used to sonoprint conductive PEDOT-S deep in tissue at biocompatible temperatures and pressures.

Discussion

In this study, we have extended sonoprinting to the domain of conductive polymers. We developed temperature-based and pressure-based strategies to sonoprint conductive polymers and

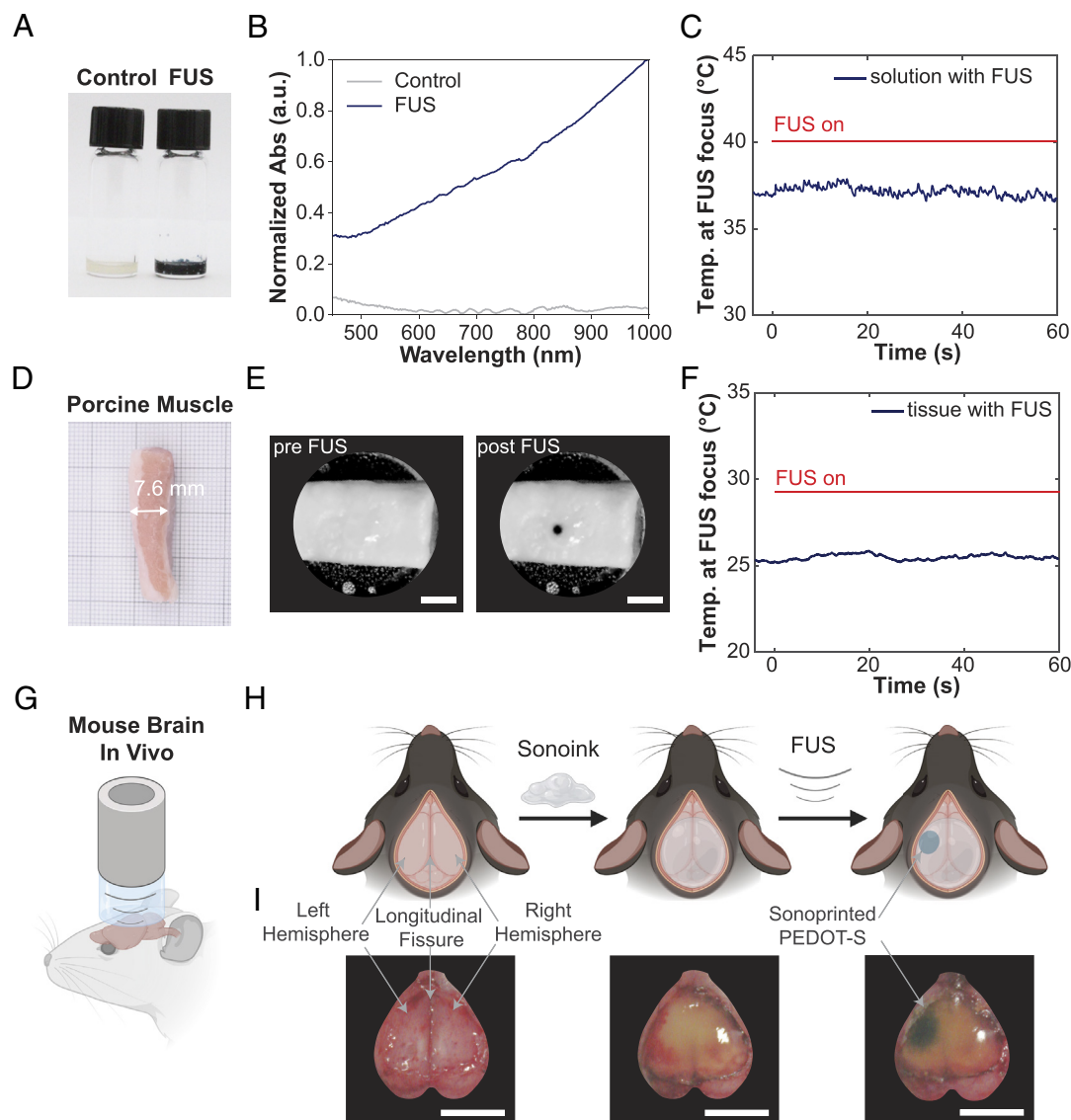


Fig. 4. Polymerization of PEDOT-S with a pressure-based sonoink in solution and tissue. (A and B) The pressure-based sonoink was polymerized in an aqueous solution by applying FUS. Images (A) and absorbance spectra (B) of the sonoink without applying FUS (control) and after applying FUS for 10 min (FUS). (C) Temperature of the solution at the focus exhibited negligible increase while applying FUS. (D and E) The pressure-based sonoink was polymerized through porcine muscle with a thickness of 7.6 mm (D) by applying FUS, which printed a dot on the tissue (E). (Scale bar, 5 mm.) (F) Temperature of the tissue at the focus was stable while applying FUS. FUS parameters are 5 MHz, 3.6 MPa, 1% duty cycle, and 1 s cycle length in panels A–F. (G) In vivo sonoprinting setup. The FUS transducer was positioned above a mouse brain with ultrasound gel at the interface for acoustic coupling. (H and I) Schematics (H) and white-light images (I) of the mouse brain after the craniotomy (Left), after applying the pressure-based sonoink in an agarose gel (Middle), and after printing PEDOT-S with FUS (Right). (Scale bar, 5 mm.) FUS parameters are 5 MHz, 2.7 MPa, 1% duty cycle, 1 s cycle length, and 60 s total duration in panels G–I. Schematic created in BioRender.

demonstrated that these approaches can be used to sonoprint PEDOT through optically opaque hydrogels and tissue with high spatial resolution. Furthermore, we have shown that our pressure-based approach enables sonoprinting with no temperature increase, resolving a key limitation of previous sonoprinting approaches that constrains their use for biological applications (14, 16).

Our results highlight the potential of sonoprinting for the fabrication of bioelectronic interfaces in tissue (15). Although our work focused primarily on developing methods for sonoprinting conductive polymers rather than their biological applications, a number of recent studies have explored applications of polymerizing conductive polymers in vivo (34–37). For example, a genetically targeted approach to synthesize conductive polymers on cell membranes has been used to modulate neuronal activity in mouse brain slices and behavior in *Caenorhabditis elegans* (34). Similarly, a metabolite-dependent polymerization approach has been used

to electrically stimulate leech nerves through conductive polymer electrodes (37). Our results complement this body of work by providing a method to spatially pattern conductive polymers in tissue. In future studies, this could be leveraged to target polymerization to specific brain regions or to print electrode arrays in tissue (27, 37).

The pressure-based sonoprinting approach developed here provides a flexible framework for using FUS to trigger oxidative polymerization reactions in tissue. Thus, it could be integrated with the genetically targeted or metabolite-dependent polymerization methods described above by encapsulating hydrogen peroxide, the initiator used in the genetically targeted approach, or lactate oxidase, the enzyme used in the metabolite-dependent approach, inside acoustically responsive double emulsions (34, 37).

For future bioelectronic applications, the biocompatibility of the polymerization reaction may need to be improved to minimize

cytotoxicity associated with APS, FeSO₄, and their reaction (37, 38). Toward this end, the concentrations of FeSO₄ and APS in the pressure-based sonoink were selected to be well below their respective LD50 levels [FeSO₄: 680 mg/kg in mice (39), APS: 689 mg/kg in rats (40)] with the potential of further reducing the concentration of FeSO₄ (SI Appendix, Fig. S20). Moreover, our pressure-based approach could be adapted for use with other initiator/catalyst systems that have been more extensively tested in vivo [e.g., H₂O₂/HRP as in (34, 37)] by changing the composition of the inner and outer aqueous solutions. We plan to integrate these insights into future iterations of our sonoprinting approach, with the ultimate goal of developing minimally invasive methods to fabricate bioelectronic interfaces in tissue.

Materials and Methods

Preparation of Temperature-Based Sonoink in an Alginate Hydrogel.

Alginate hydrogels with a temperature-based sonoink were prepared containing 3,4-ethylenedioxythiophene (EDOT) (MilliporeSigma, 483028-10G, Burlington, MA) and ammonium persulfate (APS) (MilliporeSigma, A3678-100G, Burlington, MA). Notably, the negatively charged alginate functions as the counterion for PEDOT, thereby eliminating the need for poly(styrenesulfonic acid) (PSS) (41). First, 78 mg of EDOT was added to 3 mL of deionized water, then sonicated for 2 min with a probe sonicator (Fisher Scientific, FB120, Waltham, MA). Next, 120 mg of sodium alginate (MP Biomedicals, 218295, Irvine, CA) was added to the EDOT solution and vortexed until completely dissolved. Separately, 120 mg of APS was dissolved in 3 mL of 10 mM CaCl₂ solution (Fisher Scientific, C79500, Waltham, MA). Finally, the APS solution and EDOT solution were combined, vortexed briefly, and poured into a mold. For circular samples, a Falcon Petri dish (35 mm diameter, 10 mm height, Corning, 351008, Corning, NY) was used as the mold. For square samples, a Tissue-Tek Cryomold (15 mm length, 5 mm height, Sakura, 4566, Torrance, CA) was used as the mold. For sonoink in tissue samples, a divot was carved into the tissue before pouring the solution. The hydrogels were then stored at 4 °C to mitigate decomposition of the APS. To obtain uniform, tightly crosslinked hydrogels, three CaCl₂ solutions of increasing concentration (50 mM CaCl₂, 200 mM CaCl₂, 800 mM CaCl₂) were added to the mold. Each CaCl₂ bath lasted ~1 h, after which it was replaced by a more concentrated CaCl₂ solution. The final concentrations of the temperature-based sonoink components were 13 mg/mL EDOT, 20 mg/mL alginate, and 20 mg/mL APS.

Preparation of Temperature-Based Sonoink in a PVA Hydrogel. Polyvinyl alcohol (PVA, MW 146,000 to 186,000) (MilliporeSigma, 363065, Burlington, MA) hydrogels with a temperature-based sonoink were prepared containing EDOT, PSS (Polysciences, 08772, Warrington, PA), and APS. First, EDOT and PSS were dissolved in an 80% dimethyl sulfoxide (DMSO) (Fisher Scientific, BP231-1, Waltham, MA), 20% deionized water solution to reach final concentrations of 15 mg/mL EDOT, and 20 mg/mL PSS, respectively. Next, PVA was added to the solution to reach a concentration of 200 mg/mL. The mixture was heated to 120 °C for 2 h, then poured into a metal mold (20 mm width, 25 mm length, 10 mm height), and placed in the -20 °C freezer for 16 h. After 16 h, the hydrogels were soaked in a solution of 50 mg/mL APS, 15 mg/mL EDOT, 20 mg/mL PSS in deionized water for 2 h to allow APS to diffuse into the hydrogels prior to printing. The final concentrations of the temperature-based sonoink components were 15 mg/mL EDOT, 20 mg/mL PSS, 200 mg/mL PVA, and 50 mg/mL APS.

FUS Printing Setup. A 5 MHz FUS transducer (Precision Acoustics, TXH-5-19, Dorchester, Dorset, UK) was connected to a three-axis motorized translation stage and placed in a water bath. Samples were placed on top of a thin plastic membrane (Kirkland Signature, stretch-tite plastic food wrap, Issaquah, WA) in contact with the surface of the water. The transducer was submerged in water with its face oriented upward and focused on the top surface of the sample. FUS printing parameters (i.e., peak pressure, duty cycle, and cycle length) are specified in figure captions.

Polymerization Kinetics of the Temperature-Based Sonoink in an Aqueous Solution. To explore the relationship between reactant concentration, temperature, and polymerization rate, kinetics measurements were performed with a temperature-controlled UV-Vis spectrophotometer (Thermo Fisher Scientific,

Evolution 350, Waltham, MA). First, EDOT and PSS were dissolved in deionized water to the desired concentrations, placed in a 1 cm cuvette in the spectrophotometer, and then heated to the desired temperature. Next, a separate solution of APS dissolved in deionized water to the desired concentration was prepared. To initiate the reaction, the APS solution was added to the EDOT/PSS solution with a volume ratio of 1:4. Absorbance spectra were measured with a wavelength range of 190 to 1,100 nm, data interval of 1 nm, bandwidth of 4 nm, and scan speed of 2,000 nm/min. Temperature in the cuvette was controlled with a custom temperature controller, and the solution was stirred with a magnetic stir plate throughout the reaction. The final concentrations of EDOT, APS, PSS, and temperatures for each reaction are shown in SI Appendix, Table S1. Because PSS functions as a surfactant for solubilizing EDOT and does not participate in the reaction, PSS concentrations were scaled proportionally to EDOT (42).

Microfluidic Device Fabrication. Details of the microfluidic device design and fabrication have been previously reported (32). In short, photolithography was used to generate molds for the device on a silicon wafer. The mold and a blank silicon wafer were filled with polydimethylsiloxane (PDMS) precursors (MG Chemicals, RTV615, Burlington, Ontario, Canada), then baked at 80 °C for 12 to 15 min. After removing the PDMS from the mold, a 1 mm biopsy punch (Robbins Instruments, RBP-10, Sunnyvale, CA) was used to create holes for the inlet and outlet ports, and Scotch tape was used to clean the PDMS pieces. Finally, the PDMS slabs from the mold and the blank wafer were joined together and baked at 80 °C for 48 to 56 h. CAD files for the device have been made available in a public repository at the following link: <https://osf.io/h4sr9/>.

Double Emulsion Preparation. Water/perfluorohexane/water double emulsions with a perfluorohexane (PFH) oil shell and an aqueous core with APS were produced with a flow focusing microfluidic device (32, 43, 44). First, the solutions comprising the aqueous phases and oil phase of the double emulsions were prepared. Next, the microfluidic device was plasma-treated to enhance the hydrophilicity of the section of the device from the outer aqueous inlets to the outlet port. Finally, solutions were injected into the device to produce double emulsions. Details of solution preparation, plasma treatment, and flow conditions are provided below.

Solution Preparation. The solutions used to produce the double emulsions were prepared as follows.

Oil solution. The oil solution was prepared by dissolving 40 mg of Honey dSurf (Fluigent, ODSURF5G, Le Kremlin-Bicêtre, Île-de-France, France) in 2 mL of PFH (Millipore Sigma, 379247-50ML, Burlington, MA). The solution was then loaded into a 5 mL syringe (BD, 309646, Franklin Lakes, NJ).

Inner aqueous solution. The inner aqueous solution was prepared with a buffer of 1% Tween 20 (Fisher BioReagents, BP337, Waltham, MA) in phosphate-buffered saline (PBS), pH 7.4 (Gibco, 10010031, Waltham, MA). 1.16 g of APS was dissolved in 2 mL of the buffer to provide a source of free radicals for polymerization, and 4 µL of a 20 mM fluorescein (Millipore Sigma, 46955, Burlington, MA) stock solution was added for visualization. The solution was then filtered with a 0.22 µm Millex PVDF syringe filter (Millipore Sigma, SLGVR33RB, Burlington, MA), and loaded into two 1 mL syringes (BD, 309628, Franklin Lakes, NJ).

Outer aqueous solution. The outer aqueous solution was prepared with a buffer solution of 1% Tween-20 and 2% Pluronic F-68 (Millipore Sigma, K4894, Burlington, MA) in PBS. 2 g of NaCl (Thermo Fisher Scientific, S271-500, Waltham, MA) were dissolved in 10 mL of the buffer to balance the osmolarity of the inner solution. The solution was then filtered with a 0.22 µm Millex PVDF syringe filter and loaded into a 10 mL syringe (BD, 309603, Franklin Lakes, NJ).

Plasma Treatment. Tape was placed over the inner aqueous and oil phase inlets of the device (shown in Fig. 3A), so that the channels before the center loop (the inner aqueous and oil channels) remain hydrophobic after plasma treatment. The device was then placed in a plasma cleaner (Harrick Plasma, PDC-001, Ithaca, NY) and treated for 10 min on high RF power at a pressure of 300 to 500 millitorr of ambient air.

Flow Conditions. The inner aqueous, oil, and outer aqueous syringes were mounted on four separate syringe pumps (Harvard Apparatus, Pump 11 Pico Plus Elite, 70-4511, Holliston, MA) and connected to the device via medical tubing (Scientific Commodities, Micro Medical Tubing, 0.015" I.D. × 0.043" O.D.,

BB31695-PE/2, Lake Havasu City, Arizona). At the start, all lines were disconnected from the device. Initial flow conditions for the syringe pumps were set to 3,000 $\mu\text{L/h}$ for the outer aqueous pump, 350 $\mu\text{L/h}$ for the oil pump, and 100 $\mu\text{L/h}$ for each inner aqueous pump. Immediately after plasma treatment, the outlet line was connected to the device, and the outer solution was injected through the corresponding inlet port (Fig. 3A). After 1 to 2 min, the oil solution was injected through the oil inlet port. Solution flow was visualized with an astronomy camera (ZWO, ASI174MM, Suzhou, Jiangsu Province, China) attached to a microscope (AmScope, Irvine, CA). Once the oil had wetted the microchannel, the oil flow rate was reduced to 200 $\mu\text{L/h}$ and the inner aqueous solution was injected into the device through the corresponding inlets (shown in Fig. 3A). The flow rates were slowly adjusted to final rates of ~ 125 $\mu\text{L/h}$ for the oil pump, ~ 28 $\mu\text{L/h}$ for each inner pump, and $\sim 3,500$ $\mu\text{L/h}$ for the outer pump. Once the flow conditions stabilized, double emulsions were collected for all experiments in microcentrifuge tubes for 10 min per tube. The double emulsions were then stored at 4 $^{\circ}\text{C}$, and subsequent experiments were performed within two weeks of emulsion generation to minimize decomposition of the initiator.

Double Emulsion Imaging and Size Quantification. Images of double emulsions (Fig. 3C) were acquired with a camera (Leica Microsystems, MC170 HD, Wetzlar, Hesse, Germany) attached to a light microscope (Leica Microsystems, DM2700 M, Wetzlar, Hesse, Germany) with 10 $\times$ and 50 $\times$ objectives. The diameter of the inner aqueous core and oil shell was quantified with Cellpose 1.0 (45) in images acquired with the 50 $\times$ objective.

FUS Polymerization of the Pressure-Based Sonoink in an Aqueous Solution. A monomer solution of 30 mg/mL 4-((2,3-dihydrothieno[3,4-b][1,4]dioxin-2-yl)methoxy)butane-1-sulfonate (EDOT-S) (AmBeed, A116591, Arlington Heights, IL), 30 mg/mL iron (II) sulfate heptahydrate ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$) (Thermo Fisher Scientific, 423420050, Waltham, MA), and 200 mg/mL NaCl in deionized water was prepared. Separately, double emulsions were washed with saline to remove any components of the inner aqueous solution that diffused out of the double emulsions during storage. Specifically, a microcentrifuge tube of double emulsions was centrifuged at 500 rpm for 60 s, and the supernatant was replaced with 200 mg/mL NaCl solution three times. Next, 50 μL of settled double emulsion droplets and 50 μL of the monomer solution were placed on the thin plastic membrane suspended in the water bath above the transducer. In line with previous acoustic droplet vaporization studies (31), the water bath temperature was set to 37 $^{\circ}\text{C}$ to mimic physiological conditions. The combined monomer/double emulsion solution was then treated with FUS for 5 min by a 5 MHz transducer at two different spots to ensure high vaporization efficiency. FUS parameters were 1% duty cycle, 1 s cycle length, and 3.6 MPa. During ultrasound application, solution temperature was monitored with a thermal camera (Teledyne FLIR, FLIR A300, Wilsonville, OR). After ultrasound, the absorbance spectrum was measured with a UV-Vis spectrophotometer and compared to a control sample with the same monomer/double emulsion solution placed in the 37 $^{\circ}\text{C}$ water bath for 10 min that did not receive ultrasound pulses.

FUS Polymerization of the Pressure-Based Sonoink in Tissue. The monomer and double emulsion solutions were prepared as described above. 10 μL of double emulsion solution and 10 μL of monomer solution were dispensed into a small divot in a slice of pork belly (Safeway, Pleasanton, CA) or sheep brain (Carolina Biological Supply, 228701, Burlington, NC). The divot was covered with 2 wt% agarose (Lonza Biosciences, 50101, Walkersville, MD) to prevent streaming during FUS. The sample was then treated with FUS for 10 min with the 5 MHz transducer, using the same parameters as in the solution polymerization experiment above. During ultrasound application, tissue temperature was monitored with the thermal camera, and polymerization was monitored with a digital camera (Nikon, D7200, Shinagawa-ku, Tokyo, Japan).

Conductivity Measurement of the Pressure-based Sonoink Polymerized With FUS. A monomer solution of 30 mg/mL EDOT-S and 30 mg/mL $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ in deionized water was prepared. Separately, double emulsions were washed with deionized water. Specifically, a microcentrifuge tube of double emulsions was centrifuged at 500 rpm for 60 s, and the supernatant was replaced with deionized water three times. Next, 50 μL of the double emulsion solution and 50 μL of the monomer solution were placed on the thin plastic membrane suspended in a water bath above a 650 kHz transducer (Image Guided Therapy, Pessac, France). The combined monomer/double emulsion solution was then treated with FUS

for 5 min. FUS parameters were 1% duty cycle, 1 s cycle length, and 2.7 MPa. After sonication, the solution was drop-casted onto a glass slide and dried on a 60 $^{\circ}\text{C}$ hot plate overnight (14 h) to form a rectangular film (3 mm width, 10 mm length). The conductivity of the film was measured with a four-point probe (Ossila, T2001A-0482, Sheffield, England) and compared to a commercial PEDOT (MilliporeSigma, 483095-250G, Burlington, MA) film. To prepare the commercial PEDOT film, the PEDOT solution was drop-casted onto a glass slide and dried on a 60 $^{\circ}\text{C}$ hot plate overnight (14 h) to form a rectangular film (3 mm width, 10 mm length). Film thicknesses were measured with a profilometer (KLA, Tencor Alpha Step D-500, Milpitas, CA).

Preparation of Pressure-Based Sonoink in an Agarose Hydrogel. Agarose hydrogels with a pressure-based sonoink were prepared containing EDOT-S, FeSO_4 , and double emulsions. First, 2 wt% agarose (Lonza Biosciences, 50101, Walkersville, MD) was dissolved in a 20% (w/v) sodium chloride solution (MilliporeSigma, S9888, Darmstadt, Germany) by heating at 80 $^{\circ}\text{C}$ for 10 min. Next, EDOT-S and $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ were added to the agarose solution to reach a concentration of 30 mg/mL for both components, and the solution was reheated for 10 min at 80 $^{\circ}\text{C}$ for 10 min to enhance dissolution of the components. The resulting solution was dispensed onto a thin plastic membrane over a water bath and allowed to solidify. A double emulsion solution (prepared as described above) was then deposited on top of the gel and covered with an additional layer of 2 wt% agarose (without EDOT-S and APS for visualization) to fix the droplets in the gel. A gamma correction of 1.5 was applied to all images uniformly to highlight the printed PEDOT-S dot against the background.

Vertebrate Animal Subjects. All animal procedures were approved by Stanford University's Administrative Panel on Laboratory Animal Care (APLAC) and Institutional Animal Care and Use Committee (IACUC). Mice had access to food and water ad libitum and were kept in facilities maintained for a 12 h light/dark cycle. Adult C57BL/6J mice (4 wk old) were received from the Jackson Laboratory (000664) and allowed to acclimate for at least 3 d before enrolling them in experiments.

Pressure-Based Sonoprinting of PEDOT-S In Vivo. A craniotomy was performed under anesthesia to expose the brain regions of interest, then the pressure-based sonoink in a 2 wt% agarose hydrogel (prepared as described above) was applied to the exposed tissue. The mouse was then positioned under the 5 MHz FUS transducer with ultrasound transmission gel (Parker Laboratories, Aquasonic 100, Fairfield, NJ) at the interface, and FUS was applied with 2.7 MPa, 1% duty cycle, and 1 s cycle length. Following FUS application, the gel and transducer were removed, and an image of the resulting PEDOT-S dot was acquired.

Preparation of FUS-Polymerized PEDOT:PSS for ^1H NMR Measurements. A solution of 13 mg/mL EDOT, 16.5 mg/mL PSS, and 20 mg/mL APS in D_2O was prepared. 50 μL of the solution was then transferred to a plastic microcentrifuge tube (Fisherbrand, 05-408-128, Waltham, MA) and treated with FUS for 5 min using the 5 MHz transducer with a 90% duty cycle (0.9 s on, 0.1 s off), 1 s cycle length, and 3.6 MPa. Similar to the alginate and PVA hydrogels, the plastic microcentrifuge tube generates heat under FUS sufficient to polymerize a small volume of the sonoink. This process was repeated to obtain a sufficient quantity of material for ^1H NMR measurements.

Preparation of FUS-Polymerized PEDOT-S for ^1H NMR Measurements. A monomer solution of 30 mg/mL EDOT-S and 30 mg/mL $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ in deuterium oxide (D_2O ; MilliporeSigma, 151882, Burlington, MA) was prepared. Separately, double emulsions were washed three times by centrifugation at 500 rpm for 60 s, followed by the replacement of the supernatant with D_2O . Next, 50 μL of the double emulsion solution and 50 μL of the monomer solution were combined on a thin plastic membrane suspended in a water bath above the 5 MHz FUS transducer. The mixture was then treated with FUS for 5 min with a 1% duty cycle, 1 s cycle length, and 2.7 MPa. This process was repeated to obtain a sufficient quantity of material for ^1H NMR measurements.

^1H NMR Spectroscopy. The commercial PEDOT:PSS, FUS-polymerized PEDOT:PSS, and FUS-polymerized PEDOT-S samples were then diluted with D_2O and placed in centrifuge tubes with 30 kDa filters (MilliporeSigma, Amicon Ultra UFC9030, Burlington, MA). Each sample was centrifuged (Thermo Fisher Scientific, Sorvall Legend X1R, Waltham, MA) at 4,000 RPM for 30 min, then the filtrate was discarded and an equivalent volume of D_2O was added. This centrifugation and D_2O wash

step was repeated at least two times per sample. ¹H NMR spectra were acquired on a 500 MHz spectrometer (Bruker Corporation, Neo500, Billerica, MA). The spectra were postprocessed with phase and baseline correction using the Mnova software (Mestrelab Research S.L.U., Mnova 16.0.0, Santiago de Compostela, Spain).

Comparison of Recorded ¹H NMR Spectra to Literature. The spectra for the commercial PEDOT:PSS solution (SI Appendix, Fig. S1) and for the FUS-polymerized PEDOT:PSS solution (SI Appendix, Fig. S2) are consistent with published data (46), with multiplets between 3 to 4 ppm corresponding to protons in PEDOT (doped state) and peaks between 1 to 2 ppm and 6.5 to 7.5 ppm corresponding to protons in PSS (46, 47). The spectrum for FUS-polymerized PEDOT-S (SI Appendix, Fig. S12) contains only the multiplets between 3 to 4 ppm corresponding to PEDOT because PEDOT-S does not contain PSS. Taken together, these results are consistent with the polymerization of PEDOT:PSS and PEDOT-S under ultrasound conditions.

Measurement of Ultrasound Pressure and Calculation of Mechanical Index and Power Density. For each transducer used in this study (650 kHz, 1.5 MHz, 5 MHz) the pressure at the focus was measured using a needle hydrophone (Precision Acoustics, NH0200, Dorchester, Dorset, UK) mounted on a motorized translation stage (Newport, VP-25XL-XYZR, Irvine, CA) and connected to an oscilloscope (Keysight, InfiniiVision MSOX2024A, Colorado Springs, CO). Pressure at the focus was first measured at a reference drive voltage V_{p1} (P_{vp1}), then pressures at other drive voltages V_{p2} (P_{vp2}) were calculated using $P_{vp1}/V_{p1} = P_{vp2}/V_{p2}$. The mechanical index (MI), was calculated as $MI = \frac{P}{\sqrt{f_c}}$, where P is the peak pressure and f_c is the center frequency of the ultrasound pulse. Finally, the spatial peak pulsed average intensity (I_{SPPA}) (48) at the ultrasound focus in water was calculated as

$$I_{SPPA} = \frac{1}{T} \int_0^T \frac{P^2}{\rho v} dt,$$

where T is the period of a single sinusoidal waveform applied by the transducer, P is the pressure at a particular output voltage, ρ is the density of water (998 kg/m³), and v is the speed of sound in water (1,480 m/s).

The peak pressure used in each experiment is reported in the corresponding figure caption. For the 5 MHz transducer, peak pressures that were used in polymerization experiments ranged from 2.3 to 3.6 MPa, corresponding to mechanical indices of 1.0 to 1.6, and power densities (I_{SPPA}) of 179 to 438 W/cm². For the 1.5 MHz transducer, the peak pressure used in polymerization experiments was 2.3 MPa, corresponding to a mechanical index of 1.9 and a power density (I_{SPPA}) of 179 W/cm². For the 650 kHz transducer, peak pressures used in polymerization experiments ranged from 0.2 to 0.6 MPa, corresponding to mechanical indices of 0.3 to 0.7 and power densities (I_{SPPA}) of 1 to 12 W/cm².

Data, Materials, and Software Availability. The data reported in this paper have been deposited in Zenodo (<https://doi.org/10.5281/zenodo.17422743>) (49).

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